

10 (**short**) A flexible chain of length L and mass m is resting on a set of scales. At time $t = 0$ the chain is lifted from one end at a steady speed v as shown in Fig. 7. The time-varying force required to lift the chain is $f(t)$.

(a) Find an expression for the force $f(t)$ at $t = 0$, the time at which the chain first lifts off the scales. [4]

(b) Sketch a graph of $f(t)$ vs time for the entire duration of the motion. Show also the reading on the scales for this same period. [6]

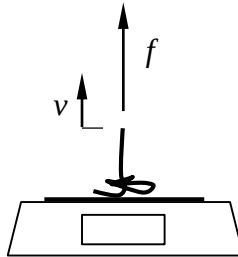


Fig. 7